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AWS Solution Architect

(UNIT-1: AWS Cloud Foundations, AWS IAM)

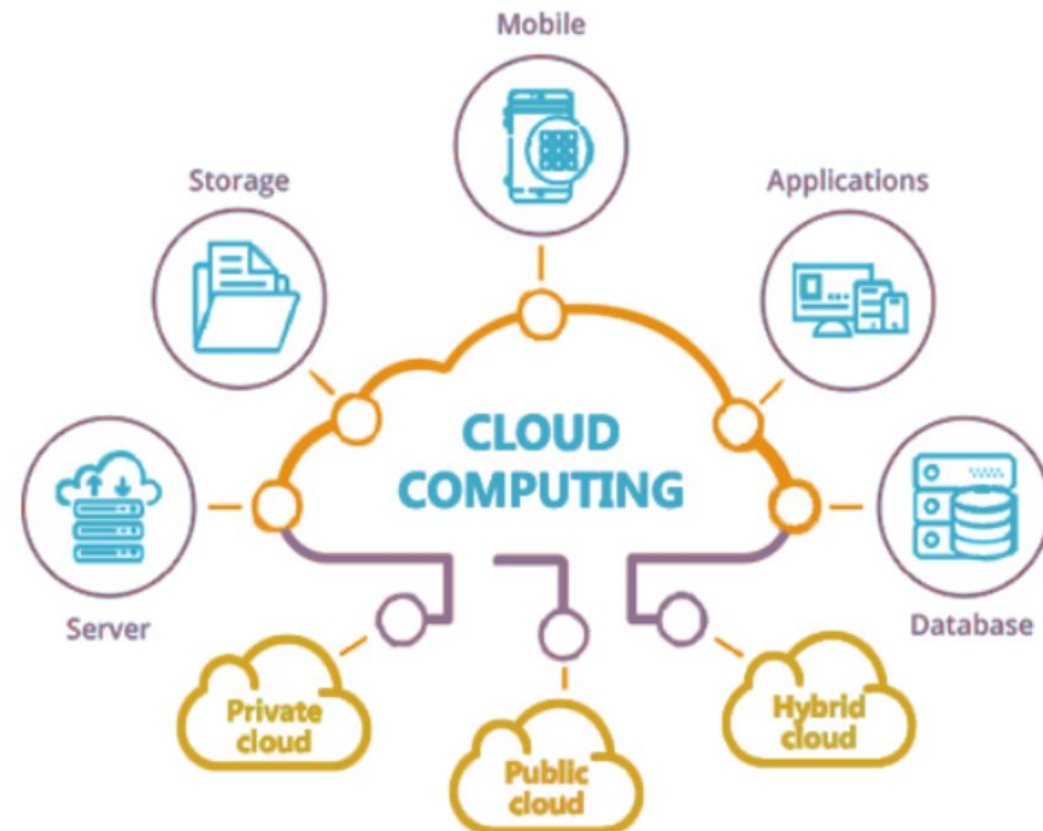
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What is Cloud Computing?



- Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing.
- Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM Cloud, Oracle Cloud, Salesforce etc.



Who is using cloud computing?



- Organizations of every type, size, and industry are using the cloud for a wide variety of use cases, such as data backup, disaster recovery, email, virtual desktops, software development and testing, big data analytics, and customer-facing web applications.
- For example, healthcare companies are using the cloud to develop more personalized treatments for patients. Financial services companies are using the cloud to power real-time fraud detection and prevention. And video game makers are using the cloud to deliver online games to millions of players around the world.

Benefits of cloud computing



Agility

- The cloud gives you easy access to a broad range of technologies so that you can innovate faster and build nearly anything that you can imagine. You can quickly spin up resources as you need them—from infrastructure services, such as compute, storage, and databases, to Internet of Things, machine learning, data lakes and analytics, and much more.
- You can deploy technology services in a matter of minutes, and get from idea to implementation several orders of magnitude faster than before. This gives you the freedom to experiment, test new ideas to differentiate customer experiences, and transform your business.

Benefits of cloud computing



Elasticity

With cloud computing, you don't have to over-provision resources up front to handle peak levels of business activity in the future. Instead, you provision the amount of resources that you actually need. You can scale these resources up or down to instantly grow and shrink capacity as your business needs change.

Cost savings

The cloud allows you to trade fixed expenses (such as data centers and physical servers) for variable expenses, and only pay for IT as you consume it. Plus, the variable expenses are much lower than what you would pay to do it yourself because of the economies of scale.

Benefits of cloud computing

Deploy globally in minutes

With the cloud, you can expand to new geographic regions and deploy globally in minutes. For example, AWS has infrastructure all over the world, so you can deploy your application in multiple physical locations with just a few clicks. Putting applications in closer proximity to end users reduces latency and improves their experience.



Traditional Computing VS Cloud Computing

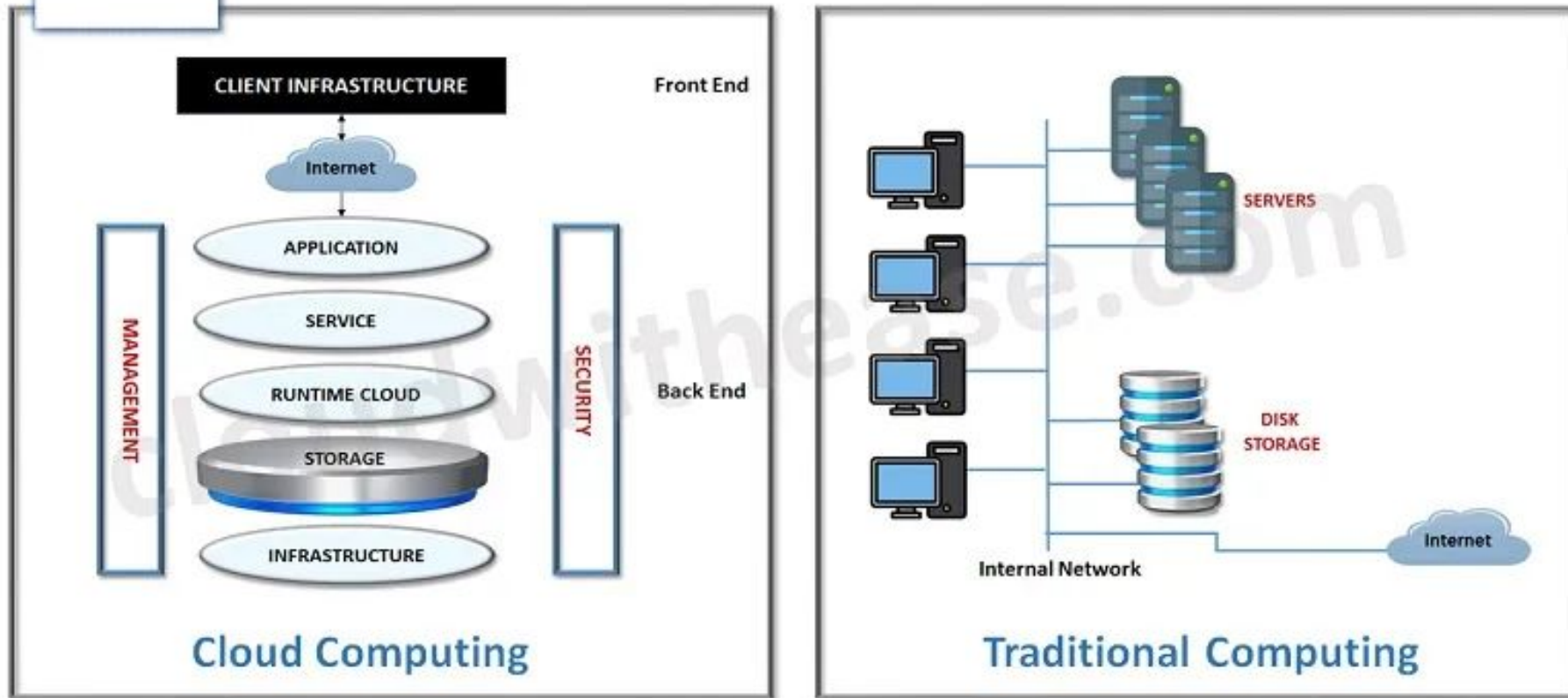


Function	Cloud Computing	Traditional Computing
Business model	Pay for use (subscription model), administrative overhead is reduced	Pay for assets, administrative costs
Concept	Delivery of different services such as data and applications though internet on different servers	Delivery of different services on local server
Data access	Ability to access data anywhere at any time by end user	User can access data only on system in which it is stored
Costs	Most cost effective as operations and maintenance of server is shared among several parties which reduces cost of public services	Less cost effective as one has to buy expensive equipment to operate and maintain server
Connectivity	Requires fast, reliable, and stable Internet connection to access data and application	Does not require Internet connectivity to access data or application
Scalability and Elasticity	It is highly scalable and elastic one can increase or decrease computing resources as per business need	Scalability and elasticity are dependent on hardware, architecture and there is a limit to scope of expansion
Resiliency and redundancy	Resiliency and redundancy are built in by cloud providers	Resiliency and redundancy levels depend on architecture, additional costs are involved to build a high resiliency architecture

Traditional Computing VS Cloud Computing



Cloud Computing vs Traditional Computing



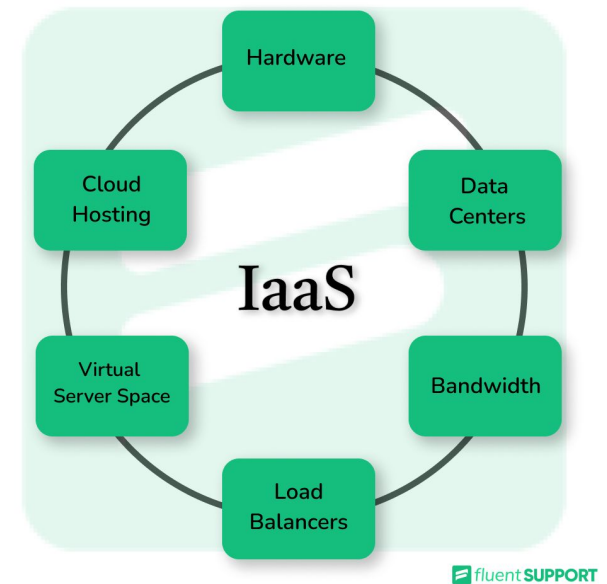
Types of cloud computing services



The three main types of cloud computing include Infrastructure as a Service, Platform as a Service, and Software as a Service. Each type of cloud computing provides different levels of control, flexibility, and management so that you can select the right set of services for your needs.

Infrastructure as a Service (IaaS)

IaaS, or Infrastructure as a Service, is a cloud computing model that provides on-demand access to virtualized computing resources like servers, storage, and networking over the internet on a pay-as-you-go basis. Instead of managing physical hardware, businesses can rent these resources, which offers greater flexibility and scalability for tasks like running applications, storing data, and development and testing environments.



Types of cloud computing services



Platform as a Service (PaaS)

PaaS, or Platform as a Service, is a cloud computing model that provides a complete platform for developing, running, and managing applications without needing to manage the underlying infrastructure, like servers, operating systems, or hardware. It offers a ready-to-use environment with development tools, databases, and other services, allowing development teams to focus on coding and deploying applications more quickly and cost-effectively.



Types of cloud computing services



Software as a Service (SaaS)

SaaS, or Software as a Service, is a software delivery model where a cloud provider hosts applications and makes them available to users over the internet through a subscription. Instead of buying and installing software locally, users access it through a web browser or thin client, while the provider handles all infrastructure, maintenance, security, and updates. This model offers greater flexibility and cost-effectiveness, as users pay for the service on a subscription basis rather than through a one-time purchase.



Types of cloud computing services

IaaS vs PaaS vs SaaS



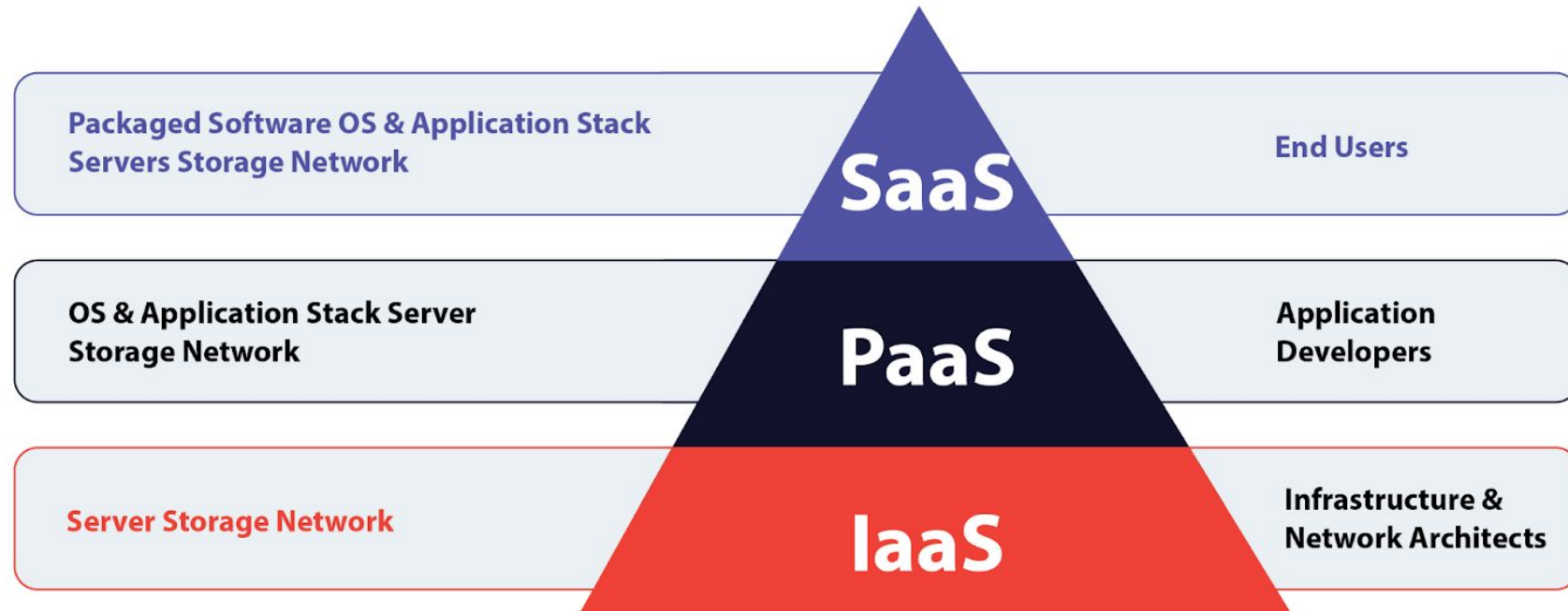
Infrastructure as a service	Platform as a Service	Software as a Service
A service model in cloud computing that provides virtualized computing resources over the internet	A cloud computing model that delivers tools necessary for application development over the internet	A service model in cloud computing that hosts software and makes them available for clients over the internet
Provides access to resources such as virtual machines, virtual storage etc.	Provides runtime environments, development and deployment tools for applications	Provides software as services to the end users
Used by network architects	Used by developers	Used by end users

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Types of cloud computing services



Cloud Service Models



Types of cloud computing services

Difference between IaaS, PaaS, and SaaS

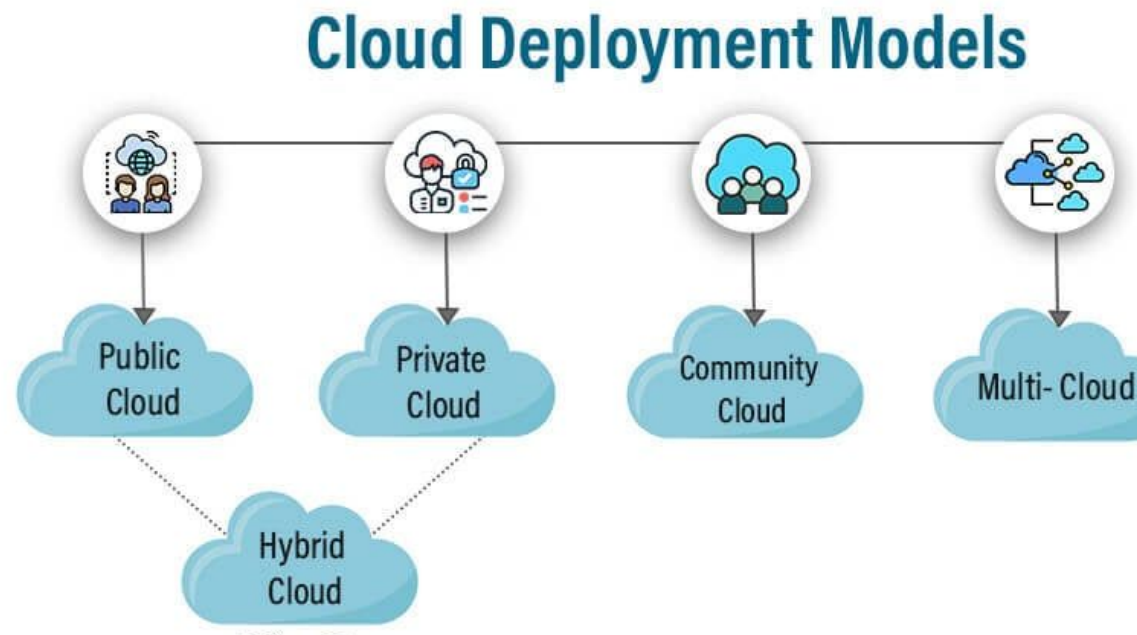


IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software.

Cloud Deployment Model



- A cloud deployment model is a specific type of cloud environment, defined by its ownership, access, and purpose, which determines how cloud resources are set up and managed for an organization. The main models are public, where resources are shared over the internet; private, where resources are for a single organization; hybrid, which combines public and private clouds; and community, where resources are shared among multiple organizations with similar needs.



What are cloud services?



- Cloud services are IT resources managed by AWS and delivered on demand over the internet. Traditionally, organizations had to purchase and configure everything from server hardware and storage systems to networking and security technologies before launching any digital system. Provisioning and managing IT infrastructure is expensive, complicated; and takes time away from innovation.
- Cloud services allow anyone to access the IT infrastructure needed to build and maintain digital systems, abstracting complex infrastructure so anyone can build sophisticated applications quickly and scale globally. Running your applications on a cloud server is just the start. You can use cloud services to add artificial intelligence and machine learning (AI/ML), real-time data analytics, and many other capabilities to your applications

What are cloud managed services?



- Cloud services are also called cloud managed services because the underlying infrastructure is fully managed by AWS. All required hardware, operating systems, and other infrastructure layers are stored and managed in highly secure AWS data centers distributed around the globe. We purchase and maintain all types of IT resources, making them available as services you can access in your application code.
- AWS monitors and maintains cloud servers, storage, networks, databases, and more, ensuring consistent performance and uptime. Beyond hardware maintenance, we handle all types of IT operational tasks from load balancing and patch management to disaster recovery and more.
- Today, thanks to AWS, anyone—from college students to enterprise teams—can access cloud services at a fraction of the cost of managing on-premises infrastructure. Anyone can build and deploy software without heavy upfront IT infrastructure investments

Introduction to AWS (Amazon Web Services)



Amazon Web Services (AWS) is one of the world's leading cloud computing platforms, providing a wide array of on-demand computing resources, tools, and services over the internet. Launched in 2006 by Amazon, AWS has become a pivotal enabler for organizations of all sizes to innovate and scale efficiently without the need for substantial upfront capital investment.

Key Features of AWS:

- **Scalability:** AWS provides elastic computing capabilities, allowing users to scale resources up or down based on demand, ensuring cost efficiency.
- **Global Infrastructure:** AWS operates in multiple regions worldwide, each with multiple availability zones, ensuring high availability and low latency.
- **Comprehensive Services:** AWS offers over 200 fully featured services, including compute (Amazon EC2), storage (Amazon S3), databases (Amazon RDS), machine learning (Amazon SageMaker), and more.
- **Pay-as-You-Go Pricing:** AWS follows a usage-based pricing model, enabling businesses to pay only for the services they use.
- **Security:** AWS prioritizes security with features like encryption, robust access controls, compliance certifications, and threat detection tools.

Introduction to AWS Cont..



Benefits of Using AWS:

- **Cost Efficiency:** AWS eliminates the need for traditional on-premises hardware and maintenance, reducing operational costs.
- **Innovation:** AWS's extensive suite of tools and services accelerates development cycles, fostering innovation.
- **Flexibility:** AWS supports a wide range of technologies, frameworks, and programming languages, catering to diverse application needs.
- **Resilience:** High reliability and disaster recovery mechanisms ensure minimal downtime and data protection.



The Beginnings of AWS: A Brief History

1. Founding and Early Days of AWS (2000–2006)

- Amazon, initially (early 2000s) focused on e-commerce, developed internal tools to manage its computing resources efficiently, enabling teams to operate independently while scaling its infrastructure to meet rapid growth.
- By 2003, Amazon realized that the infrastructure it developed for internal use could be turned into a service for external customers. This led to the idea of offering “Amazon’s Web Services” to the public, allowing other companies to use Amazon’s infrastructure to build and run their own applications.
- In 2006, AWS officially launched its first two services: **Amazon S3 (Simple Storage Service)** for scalable object storage and **Amazon EC2 (Elastic Compute Cloud)** for flexible, on-demand computing capacity. These services laid the foundation for what would become the world’s most comprehensive and widely adopted cloud platform.

2. AWS’s Early Success and Growth (2006–2010)

- This early success of Amazon S3 and EC2 encouraged AWS to expand its service offerings beyond storage and compute. Over the next few years, AWS introduced services for databases (Amazon RDS), content delivery (Amazon CloudFront), and security (AWS IAM).
- By 2010, AWS had already established itself as a major player in cloud computing, with a rapidly growing customer base that included startups, enterprises, and government organizations.

The Beginnings of AWS: A Brief History Cont.



3. Rapid Expansion of AWS Services (2010–2015)

- During this time, AWS rapidly expanded its offerings, launching services like **Amazon RDS** for managed databases, **Amazon Redshift** for data warehousing, and **AWS Lambda** for serverless computing, broadening its support for use cases such as data analytics and machine learning.
- AWS expanded globally by establishing data centers in various regions, enhancing application performance for end-users. It also launched the AWS Marketplace, allowing third-party vendors to offer complementary software and services.

4. AWS's Dominance and Market Leadership (2015-Present)

- By 2015, AWS led the cloud computing market, introducing innovations like **Amazon SageMaker** for machine learning, **AWS Lambda** for serverless computing, and IoT services, alongside strategic acquisitions and partnerships to strengthen its offerings.
- Today, AWS leads cloud computing, offering 200+ services globally. Its innovation and extensive ecosystem make it the preferred platform for businesses of all sizes.

how many regions in aws - Google x Global Infrastructure - AWS x

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Global Infrastructure Overview Regions and AZs Local Zones Wavelength Zones AWS Regional Services Sustainability Community Engagement

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AWS Global Infrastructure

The most secure, extensive, and reliable Global Cloud Infrastructure, for all your applications

Create an Account

36 launched Regions
each with multiple Availability Zones

114 Availability Zones

600+ CloudFront POPs
and 13 Regional edge caches

AWS Global Infrastructure Map

23:28

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The AWS Cloud spans 114 Availability Zones within 36 geographic regions, with announced plans for 12 more Availability Zones and four more AWS Regions in New Zealand, the Kingdom of Saudi Arabia, Taiwan, and the AWS European Sovereign Cloud.



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42 Local Zones
29 Wavelength Zones
for ultralow-latency applications

245 countries and territories served

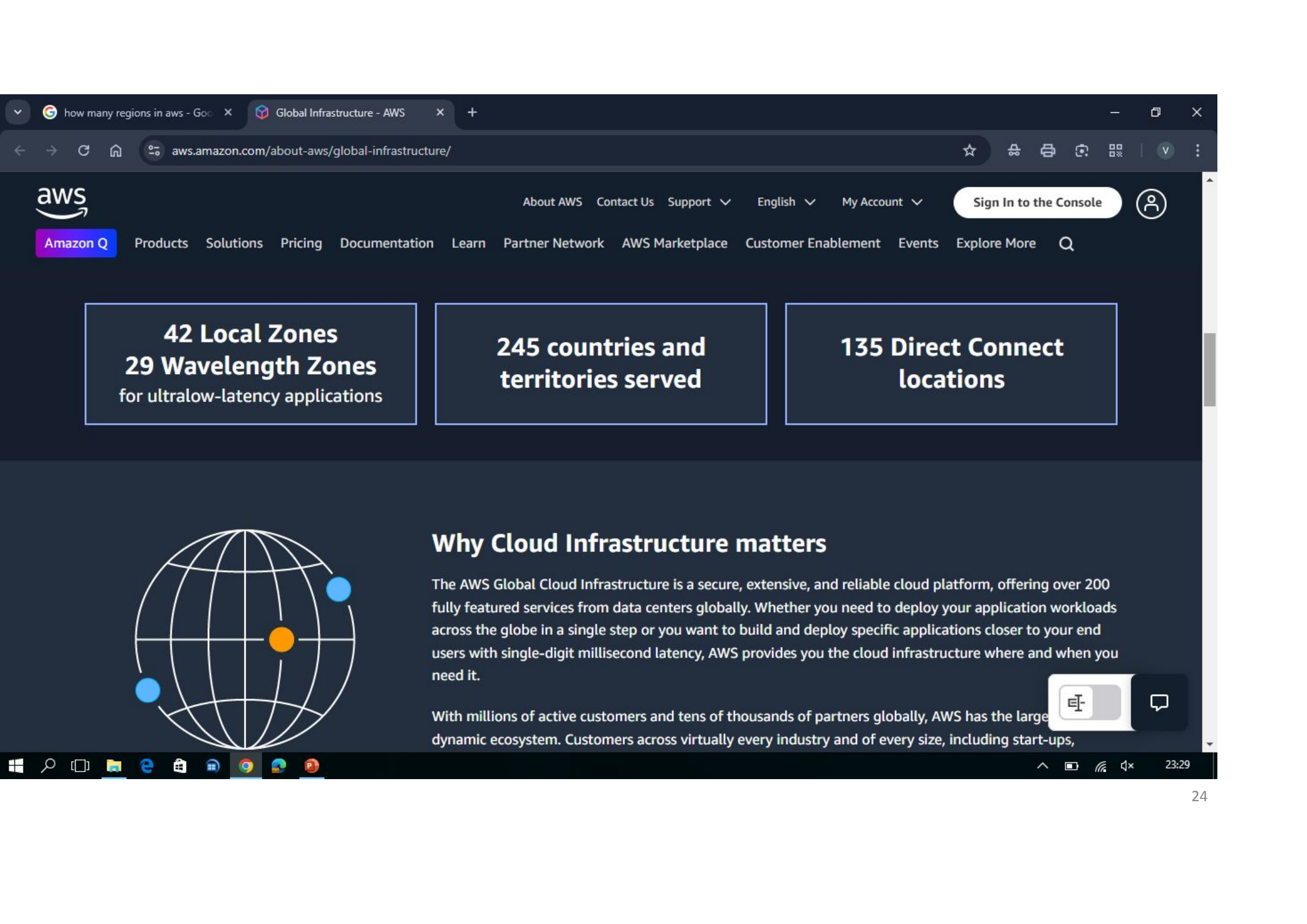
135 Direct Connect locations



Why Cloud Infrastructure matters

The AWS Global Cloud Infrastructure is a secure, extensive, and reliable cloud platform, offering over 200 fully featured services from data centers globally. Whether you need to deploy your application workloads across the globe in a single step or you want to build and deploy specific applications closer to your end users with single-digit millisecond latency, AWS provides you the cloud infrastructure where and when you need it.

With millions of active customers and tens of thousands of partners globally, AWS has the largest dynamic ecosystem. Customers across virtually every industry and of every size, including start-ups,



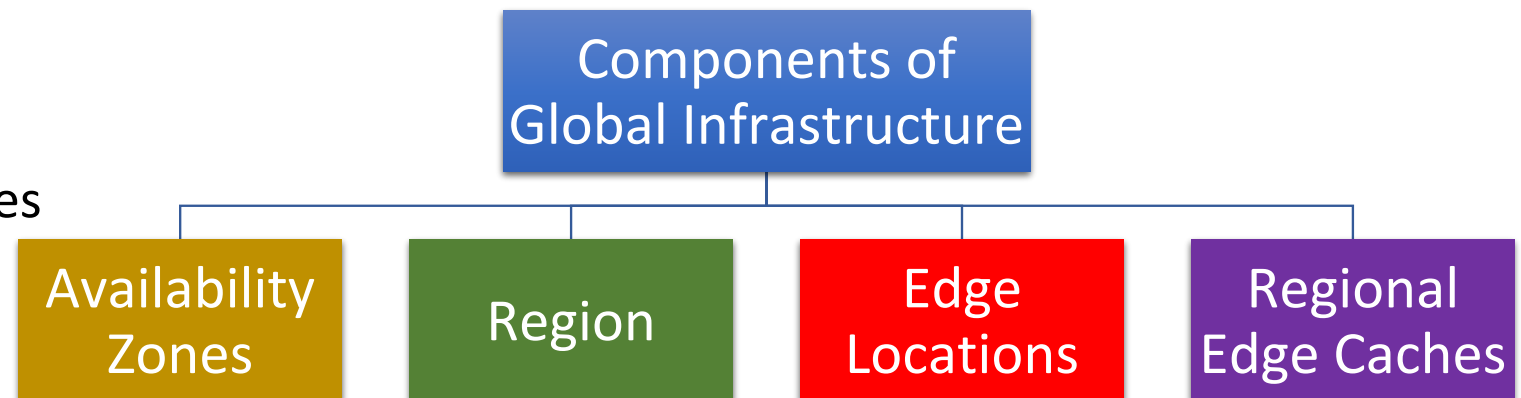
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Amazon Global Infrastructure

- AWS is a cloud computing platform which is globally available
- Global infrastructure is a region around the world in which AWS is based. Global infrastructure is a bunch of high-level IT services.

The following are the components that make up the AWS infrastructure:-

- Availability Zones
- Region
- Edge Locations
- Regional Edge Caches

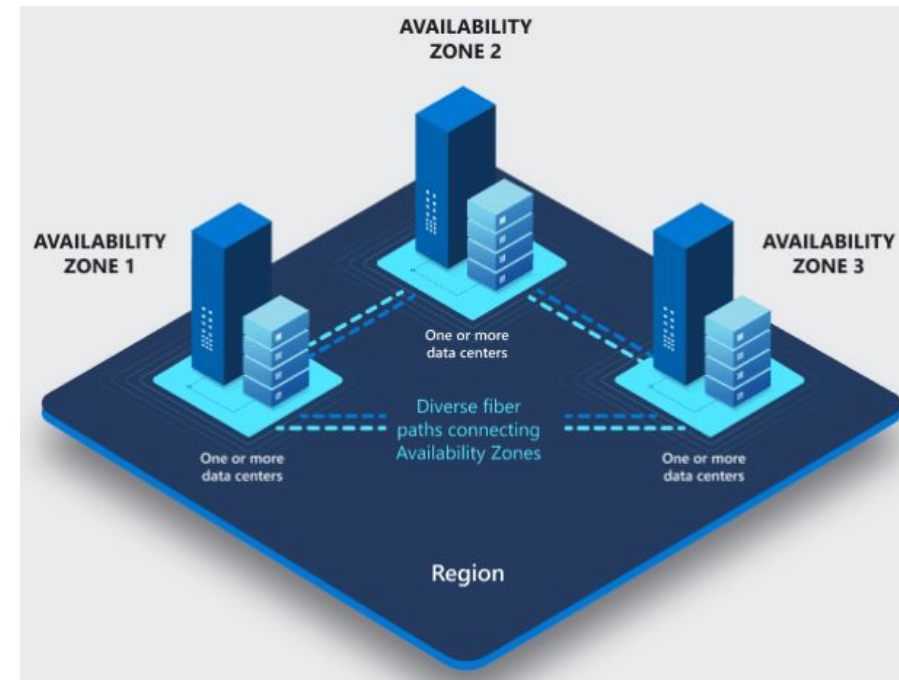


Amazon Global Infrastructure Cont.



◆ Availability Zone as a Data Center

- An AWS Availability Zone (AZ) is one or more isolated data centers within an AWS Region that have independent power, cooling, and networking. This design provides fault tolerance, so applications can remain highly available even if one AZ experiences an outage. AZs are physically separate but connected with high-bandwidth, low-latency networking to enable synchronous replication and high availability.

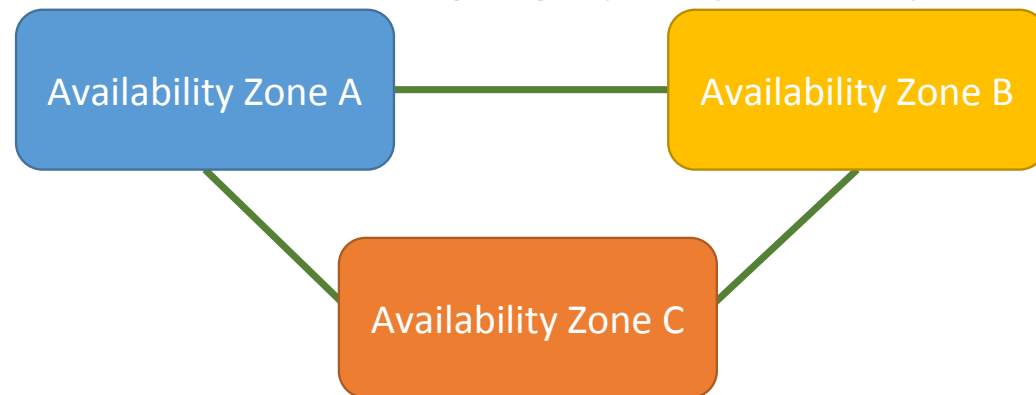


Amazon Global Infrastructure Cont.



◆ Region

- A region is a geographical area. Each region consists of 2 or more availability zones.
- An AWS Region is a physical location in the world where Amazon Web Services (AWS) clusters its data centers. Each region is a separate geographic area with multiple, isolated data centers called Availability Zones (AZs), which are connected by a high-speed, redundant network. Choosing an AWS region allows you to deploy applications and data in a location that meets specific needs for factors like geographic proximity, compliance, and latency.

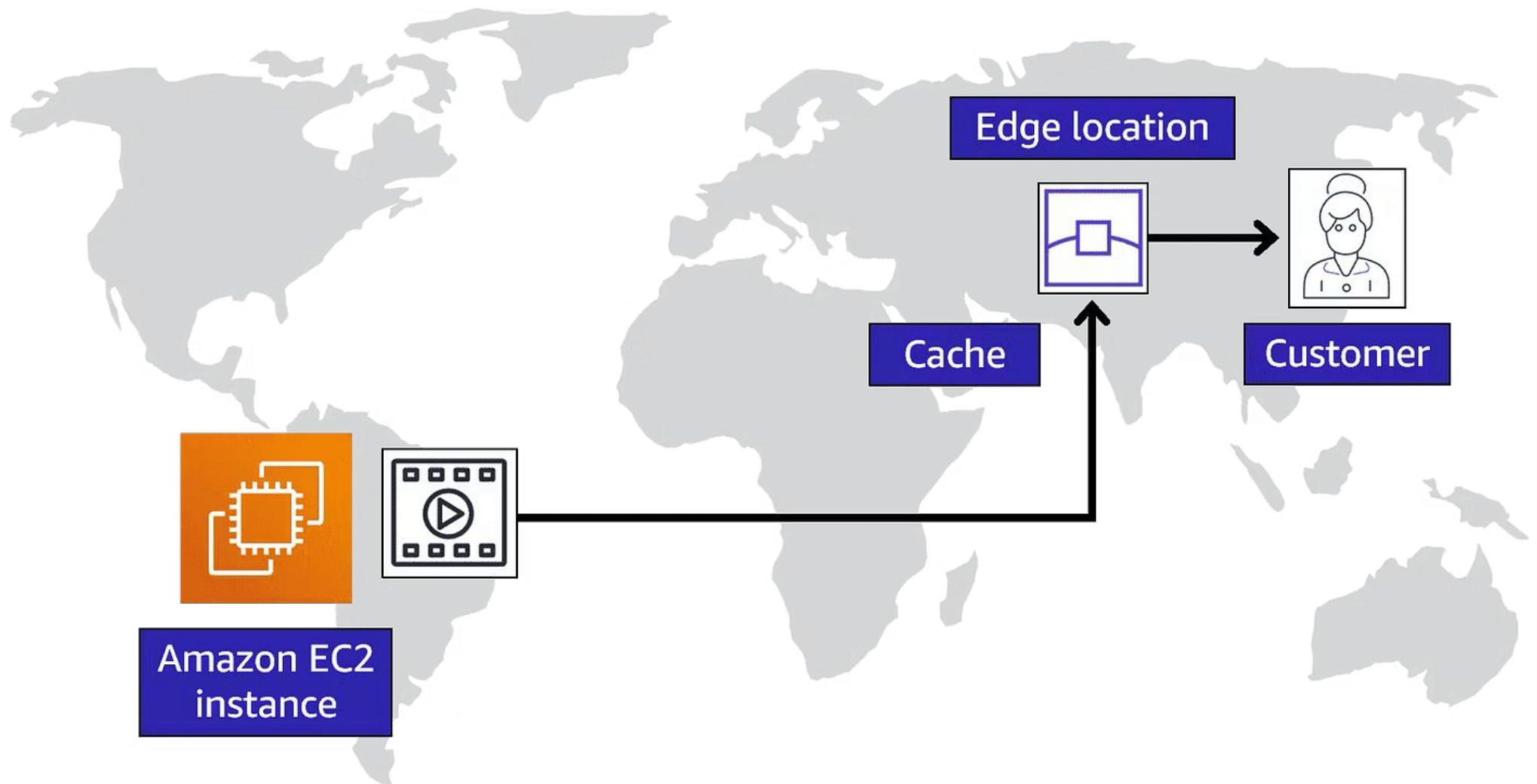


Amazon Global Infrastructure Cont.



◆ Edge Locations

- An AWS edge location is a data center located near end-users that stores cached copies of content for faster delivery.
- They are part of AWS's global network, used by services like Amazon CloudFront to reduce latency by serving content from the closest location instead of a distant origin server.
- These locations also handle tasks like request routing and network optimization.

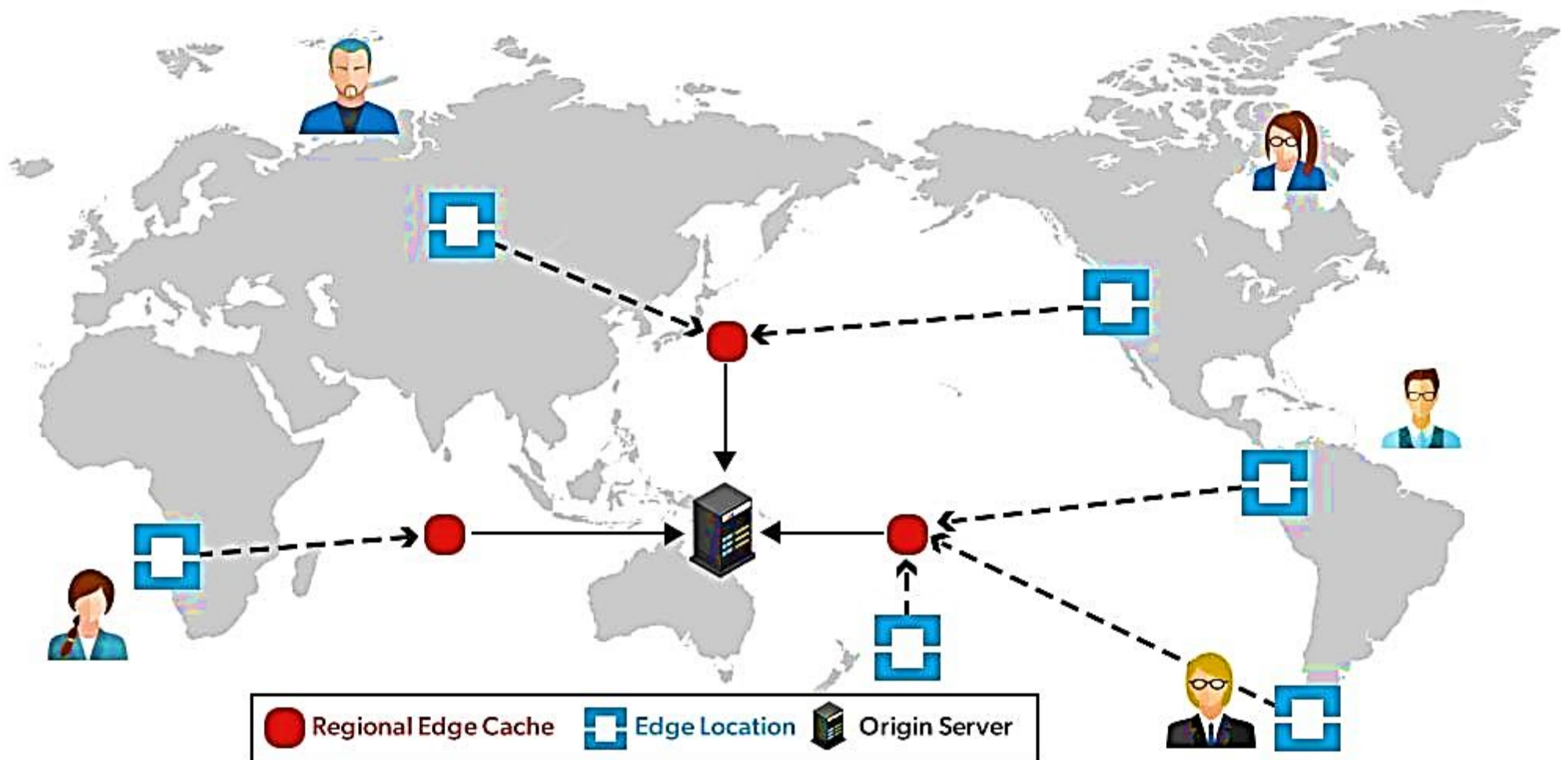


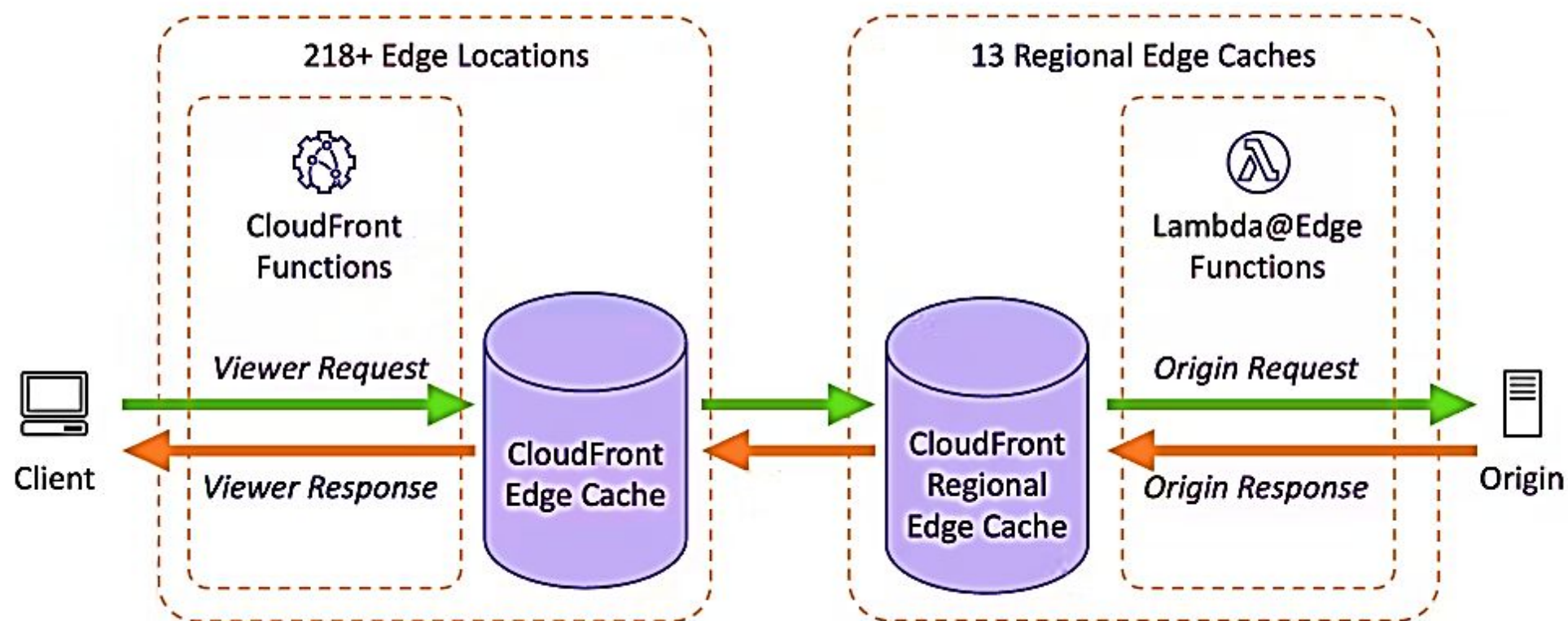
Amazon Global Infrastructure Cont.

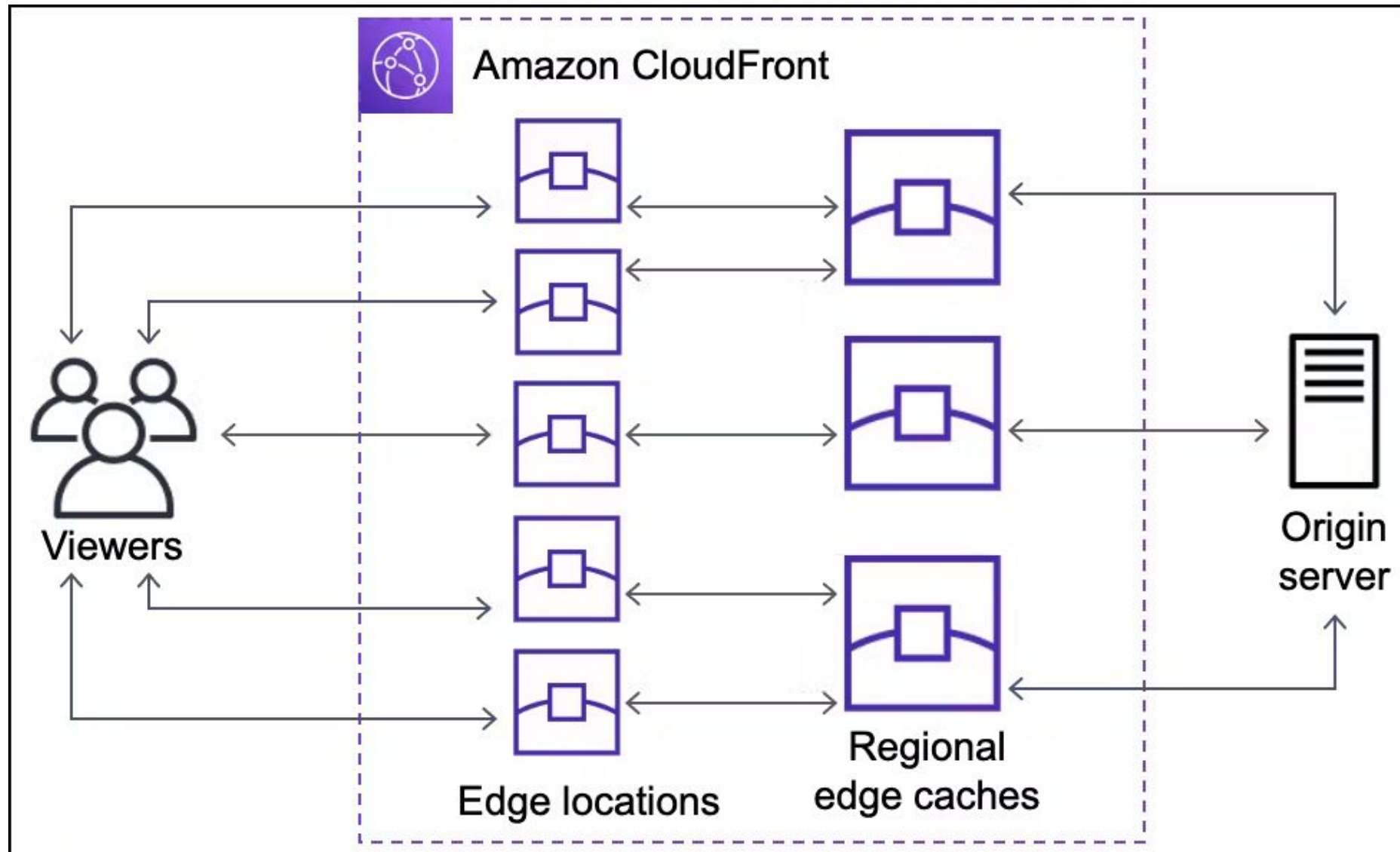


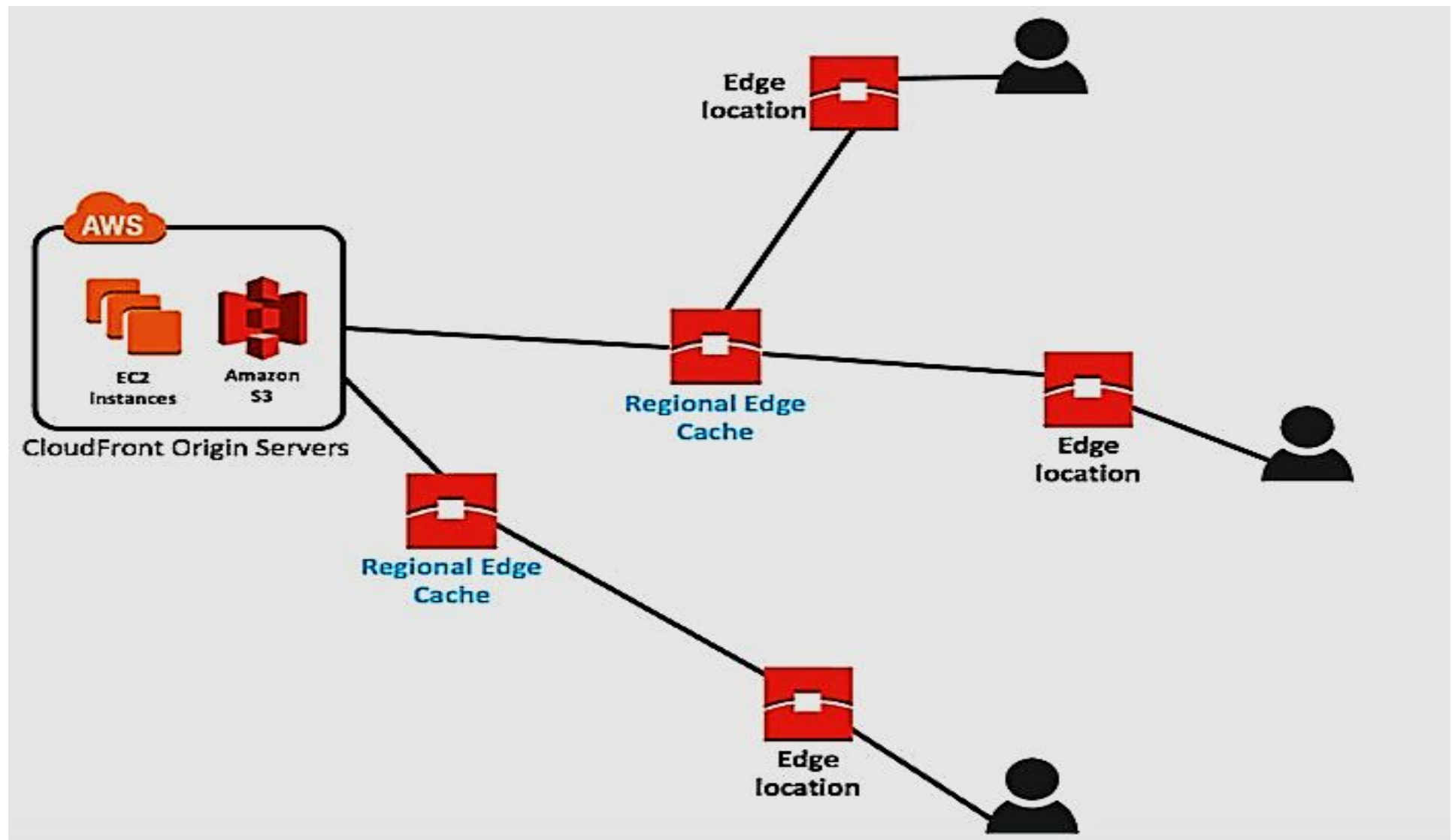
◆ Regional Edge Cache

- AWS announced a new type of edge location in November 2016, known as Regional Edge Cache Edge.
- Regional Edge Cache lies between Origin servers and the edge locations.
- A regional edge cache has a large cache than individual edge location.
- Data is removed from the cache at the edge location while the data is retained at the Regional Edge Caches.
- When the user requests the data, then data is no longer available at the edge location. Therefore, the edge location retrieves the cached data from the Regional Edge Cache instead of the origin servers that have high latency.









Amazon Global Infrastructure Cont.



◆ Concepts of Zone, Region and Multi-Region

- **Regions** are independent areas that consist of zones. They affect the pricing, reliability, networking, and performance of zonal resources - VM.
- A **zone** is a deployment area for Google Cloud resources within a region. The zone should be considered a single failure domain within a region. To deploy fault-tolerant applications with high availability and help protect against unexpected failures, deploy your applications across multiple zones in a region.
- **Multi-region services** are designed to be able to function following the loss of a single region. Multi-regional resources are cloud storage, Big Query, Big Tables, etc.

Amazon Global Infrastructure Cont.



◆ Region

- Definition: A physical geographic location that contains multiple, isolated Availability Zones. Examples include "US East (N. Virginia)" or "Asia Pacific (Tokyo)".
- Purpose: To provide redundancy and fault tolerance by separating infrastructure across a large geographic area.
- Key use: Most applications can achieve high availability by using multiple Availability Zones within a single Region.

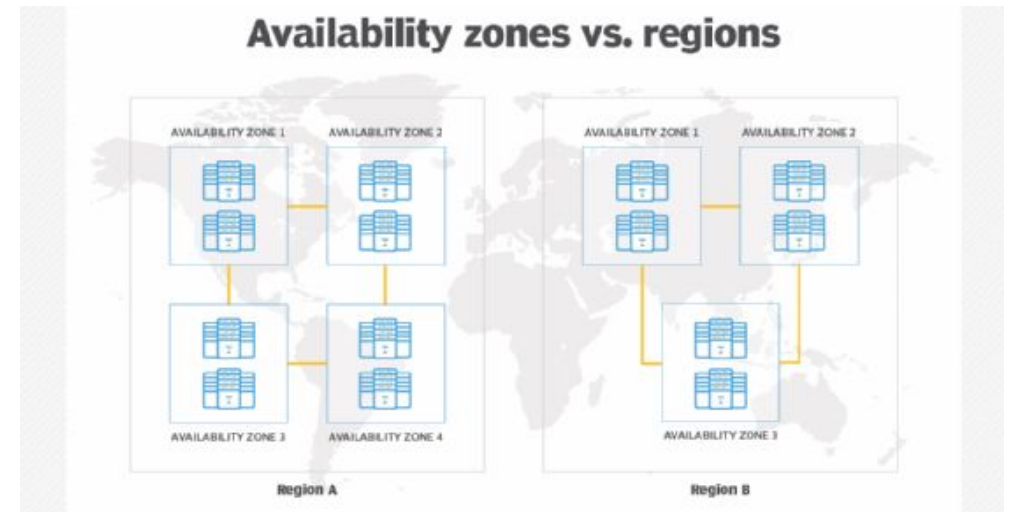


Amazon Global Infrastructure Cont.



◆ Availability Zone

- Definition: One or more discrete data centers with redundant power, networking, and connectivity, located within a Region.
- Purpose: To provide high availability and fault tolerance for applications.
- Key use: By deploying resources across multiple AZs, an application can remain available even if one AZ fails.



Amazon Global Infrastructure Cont.



◆ Multi-region

- Definition: Deploying application components across multiple AWS Regions.
- Purpose: To achieve a higher level of resilience, address disaster recovery requirements with faster recovery times, and comply with data sovereignty laws that mandate data be stored within a specific jurisdiction.
- Key use: Critical applications for a global user base or those with strict compliance needs often use a multi-Region approach to ensure business continuity and low latency.

AWS Shared Responsibility Model



The AWS Shared Responsibility Model separates security duties between Amazon and the customer. In simple terms, AWS is responsible for the security of the cloud, and the customer is responsible for security in the cloud. This division of labor helps both parties ensure a secure and compliant cloud environment.

- AWS is responsible for "security of the cloud," encompassing the infrastructure
- While customers are responsible for "security in the cloud," including their data, applications, and access controls within their AWS environment.

2. AWS Responsibilities ("Security of the Cloud"):

- **Protecting the infrastructure:** This includes the physical facilities, hardware, software, and networking that run AWS Cloud services.
- **Managing access to infrastructure:** AWS controls who can access the underlying infrastructure and resources.
- **Providing a secure platform:** AWS ensures the cloud environment is secure and reliable. 39

AWS Shared Responsibility Model



2. Customer Responsibilities ("Security in the Cloud"):

- **Managing their own data and applications:** Customers are responsible for the security of their data, applications, and the configurations of their AWS services.
- **Implementing access controls:** Customers manage who can access their resources and data within their AWS environment.
- **Configuring and patching their software:** Customers are responsible for managing the security of their guest operating systems, applications, and other software components.

3. Shared Responsibilities:

- **Controls:** Some controls are shared between AWS and the customer. For example, AWS provides the infrastructure for patching, while the customer must implement the patching process for their applications.
- **Compliance:** Both AWS and the customer are responsible for ensuring compliance with relevant regulations and standards.

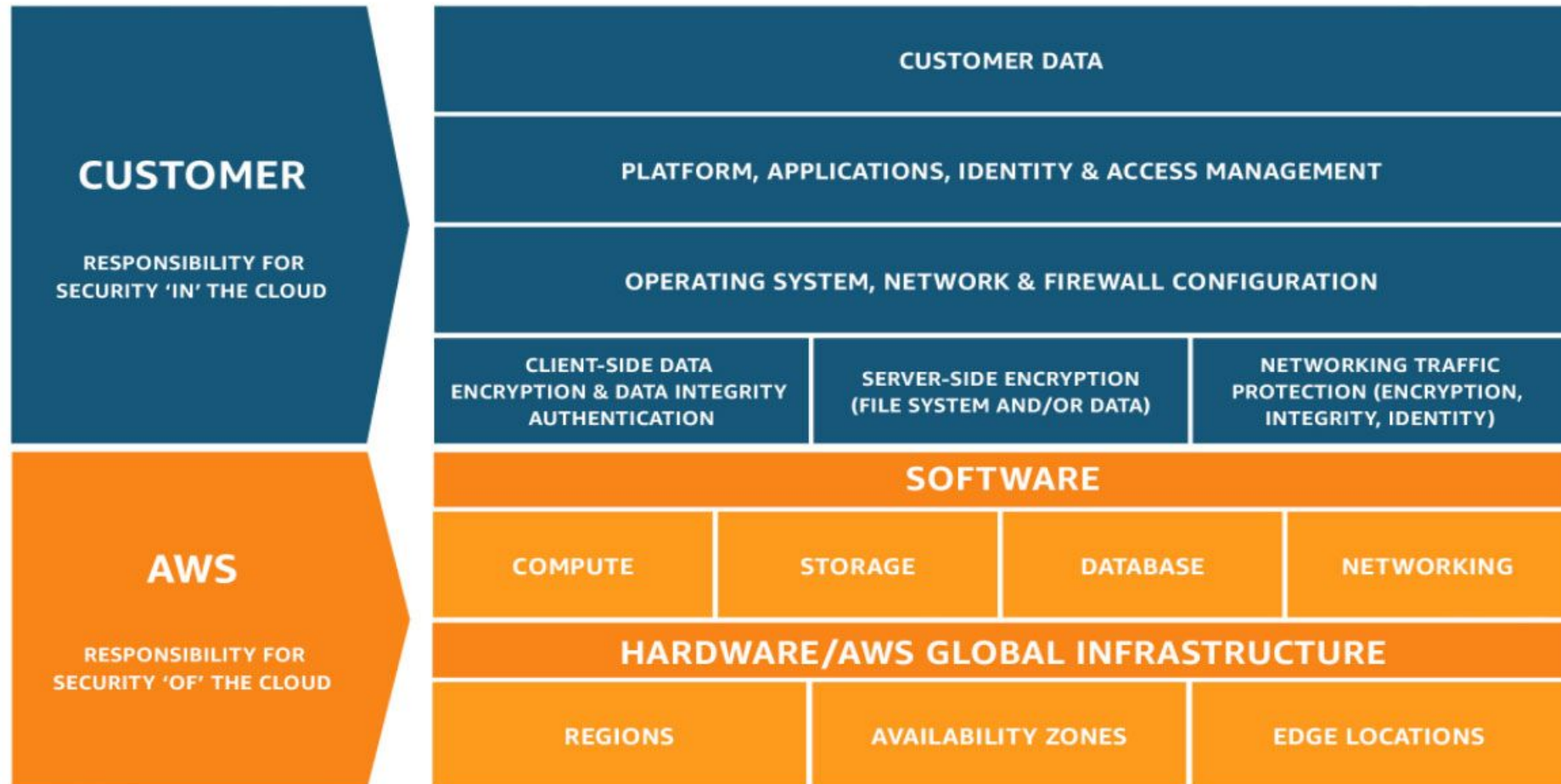
AWS Shared Responsibility Model



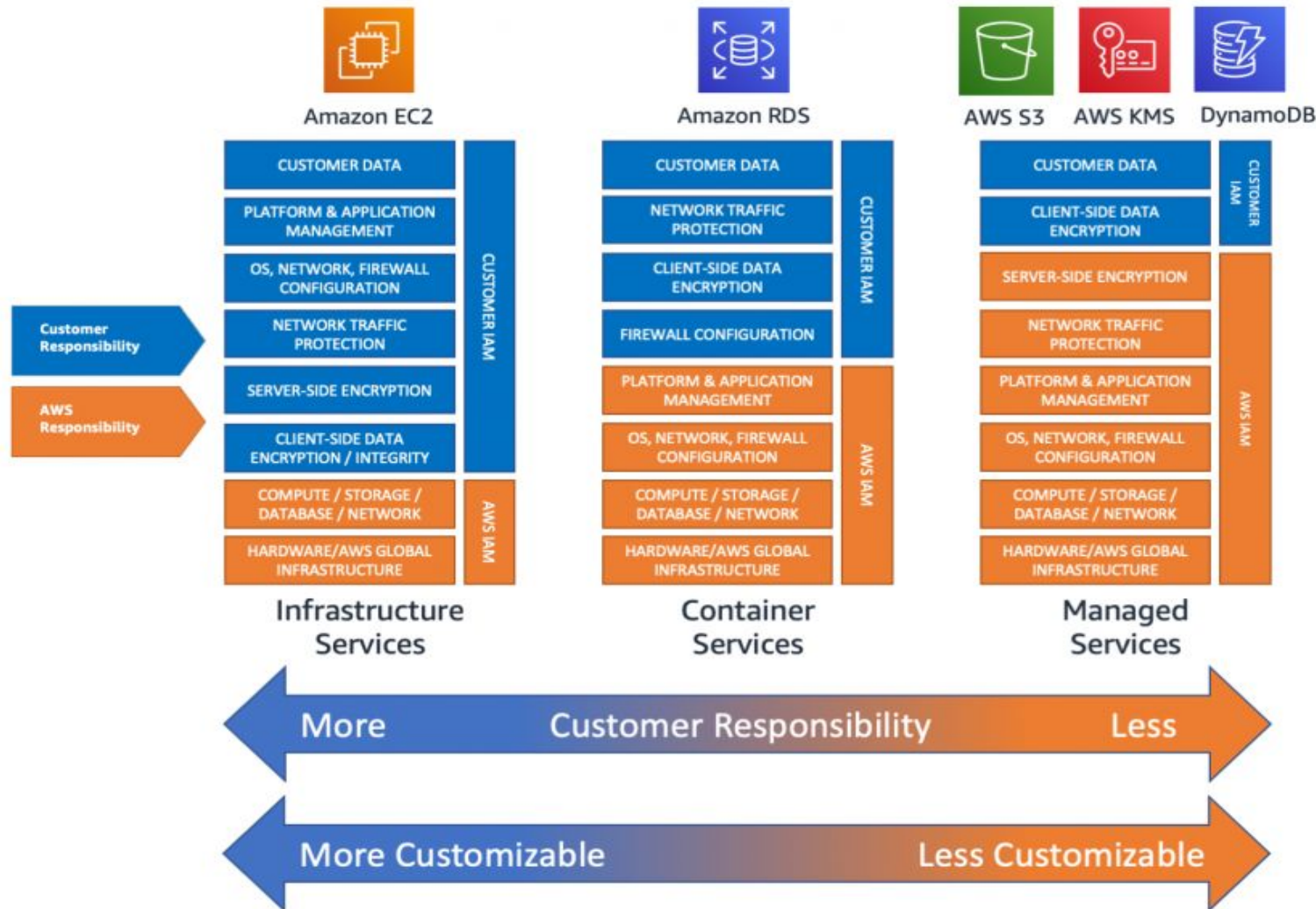
Examples:

1. **Physical security:** AWS is responsible for the physical security of the data centers, while the customer is responsible for the security of their on-premises data centers hosting AWS Outposts.
2. **Operating system:** AWS manages the security of the operating system for certain services, while customers are responsible for security configurations and patching of their chosen operating systems.
3. **Data:** Customers are responsible for encrypting their data at rest and in transit, managing access to their data, and ensuring the confidentiality and integrity of their data.

AWS Shared Responsibility Model



AWS Shared Responsibility Model



What is AWS IAM?

(Identity Access Management)



- IAM stands for Identity Access Management.
- IAM allows you to manage users and their level of access to the AWS console.
- It is used to set users, permissions and roles. It allows you to grant access to the different parts of the aws platform.
- AWS Identity and Access Management is a web service that enables Amazon Web Services (AWS) customers to manage users and user permissions in AWS.
- With IAM, Organizations can centrally manage users, security credentials such as access keys, and permissions that control which AWS resources users can access.
- Without IAM, Organizations with multiple users must either create multiple user accounts, each with its own billing and subscriptions to AWS products or share an account with a single security credential. Without IAM, you also don't have control about the tasks that the users can do.
- IAM enables the organization to create multiple users, each with its own security credentials, controlled and billed to a single AWS account. IAM allows the user to do only what they need to do as a part of the user's job.

How Does IAM Work?



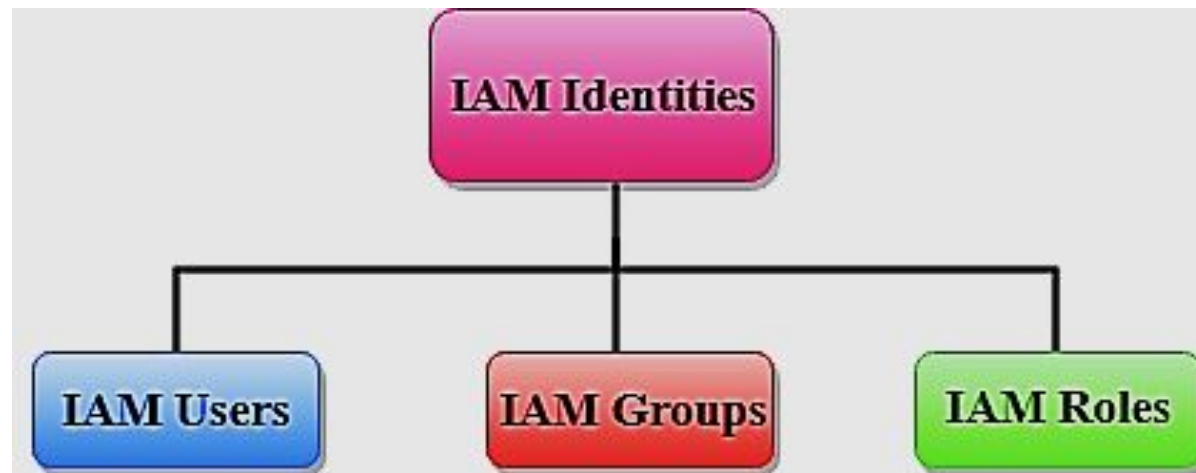
The IAM workflow includes the following six elements:

- **Principal:** A principal is an entity that can perform actions on an AWS resource. A user, a role or an application can be a principal.
- **Authentication:** It is the process of confirming the identity of the principal trying to access an AWS product. The principal must provide its credentials or required keys for authentication.
- **Request:** A principal sends a request to AWS specifying the action and which resource should perform it.
- **Authorization:** By default, all resources are denied. IAM authorizes a request only if all parts of the request are allowed by a matching policy. After authenticating and authorizing the request, AWS approves the action.
- **Actions:** Actions are used to view, create, edit or delete a resource.
- **Resources:** A set of actions can be performed on a resource related to your AWS account.

IAM Identities



- IAM identities are created to provide authentication for people and processes in your aws account.





IAM Identities contd..

IAM Users

- An IAM user is an identity with an associated credential and permissions attached to it. This could be an actual person who is a user, or it could be an application that is a user.
- With IAM, you can securely manage access to AWS services by creating an IAM username for each employee in your organization.
- Each IAM user is associated with only one AWS account. By default, a newly created user is not authorized to perform any action in AWS.
- The advantage of having one-to-one user specification is that you can individually assign permissions to each user.



IAM Identities contd..

IAM Groups

- A collection of IAM users is an IAM group.
- You can use IAM groups to specify permissions for multiple users so that any permissions applied to the group are applied to the individual users in that group as well.
- Managing groups is quite easy. You set permissions for the group, and those permissions are automatically applied to all the users in the group.
- If you add another user to the group, the new user will automatically inherit all the policies and the permissions already assigned to that group. This lessens the administrative burden.



IAM Identities contd..

IAM Groups

- An IAM (Identity and Access Management) group is a container for multiple users that lets you manage all of their permissions at once.
- Instead of setting permissions for each person individually, you can define a single set of permissions for the group. Everyone in that group then automatically gets those same permissions.

What was done previously when IAM group was not there?

- A new developer joins the company. You have to manually give them permission to access the code repository, the staging server, and the project management software. If another developer starts, you repeat this entire process.

IAM Identities contd..



IAM Policies

- An IAM policy sets permission and controls access to AWS resources.
- Policies are stored in AWS as JSON documents. Permissions specify who has access to the resources and what actions they can perform.
- For example, a policy could allow an IAM user to access one of the buckets in Amazon S3. The policy would contain the following information:
 - Who can access it
 - What actions that user can take
 - Which AWS resources that user can access
 - When they can be accessed

IAM Identities contd..

IAM Policies

In JSON
format:

```
{
  "Version": "2017-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [{
    "Sid": "AddPublicReadPermissions",
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:*",
    "Resource": ["arn:AWS:s3:::bucket/*"]
  }]
}
```

Specify Actions(Read/Write/Delete)

Give permissions(Allow/Deny)

Who can Access it

What action can a user take

Specify the resource



IAM Identities contd..



IAM Policies

There are two types of policies: **managed policies** and **inline policies**.

- Managed policies are reusable, standalone policies that can be attached to multiple users, groups, or roles, while inline policies are embedded directly into a single IAM entity (like a user or role) and are not reusable.
- A key difference is their lifecycle: inline policies are deleted when the entity they are attached to is deleted, whereas managed policies exist independently and can be attached to different entities.

IAM Identities contd..



IAM Roles

- An IAM role is like a virtual ID card that grants temporary access and permissions to resources, without needing a permanent username or password.
- You can give this "card" to different entities, such as other AWS services (like a computer needing to access a file storage), applications, or even temporary users, who can then use the role's permissions to perform specific actions.
- The key benefit is that the permissions are temporary and created on the fly, which is more secure than embedding long-term credentials directly into an application.

IAM Identities contd..



IAM Features

- IAM features are the tools and processes that manage digital identities and control who can access what in an organization's systems.
- Key features include authentication (verifying identity), authorization (granting permissions), single sign-on (SSO) for simplified access, multi-factor authentication (MFA) for enhanced security, and audit logs to track user activity.
- **Shared Access to your Account:** A team working on a project can easily share resources with the help of the shared access feature.
- **Free of cost:** IAM feature of the Aws account is free to use & charges are added only when you access other Amazon web services using IAM users.
- **Have Centralized control over your Aws account:** Any new creation of users, groups, or any form of cancellation that takes place in the Aws account is controlled by you, and you have control over what & how data can be accessed by the user.
- **Grant permission to the user:** As the root account holds administrative rights, the user will be granted permission to access certain services by IAM.
- **Multifactor Authentication:** Additional layer of security is implemented on your account by a third party, a six-digit number that you have to put along with your password when you log into your accounts.